



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2013
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9648/03

Applications Paper and Planning Question

23 September 2013

Paper 3

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough workings.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions.

At the end of the examination,

1. Fasten all your work securely together.
2. Hand in the following separately:
 - Structured questions 1 – 3
 - Planning question
 - Essay question

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
1	/ 15
2	/ 16
3	/ 9
4 Planning	/ 12
5 Essay	/ 20
Total	/ 72

This paper consists of **11** printed pages.

[Turn over]

Answer **all** questions.

For
Examiner's
Use

QUESTION 1

pCMV6-XL5 is a plasmid with a multiple cloning site (MCS) that lies downstream of the CMV promoter (Fig. 1.1). This plasmid can be inserted into both eukaryotic and prokaryotic host cells. The arrows denote the direction in which the genes are transcribed.

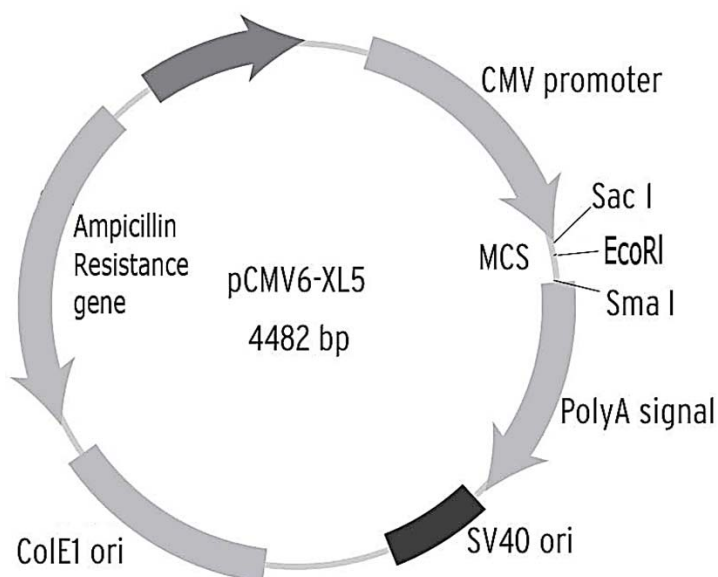


Fig. 1.1

(a) State the function of the CMV promoter in pCMV6-XL5.

[1]

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An artificially-synthesised human growth hormone (hGH) gene with flanking *Hind*III restriction site sequences was created. The restriction sites for the restriction enzymes *Hind*III and *Sac*I are shown in Fig. 1.2.

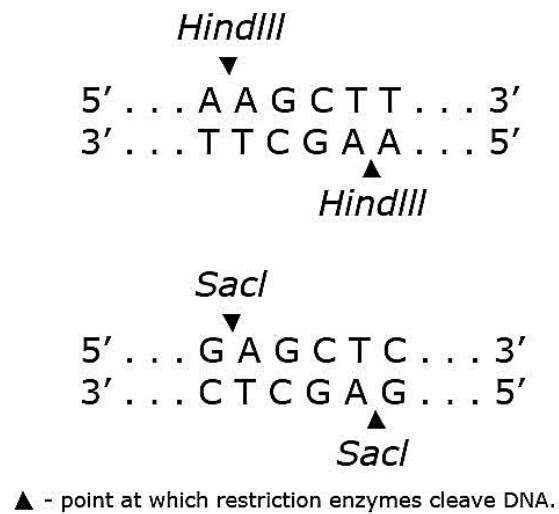


Fig. 1.2

- (b)** Using information in Fig. 1.1 and 1.2, explain how the hGH gene can be inserted into pCMV6-XL5. [3]

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- (c)** State the conditions required for artificial transformation of *E. coli* cells with the plasmid vector. [2]

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- (d)** Describe the steps required to select bacterial colonies containing the human growth hormone gene after artificial transformation has taken place. [5]

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- (e)** Suggest an explanation for the inclusion of the polyA signal sequence in pCMV6-XL5. [2]

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- (f)** Plasmid replication is usually initiated by RNAll, which is a sequence of ribonucleotides that binds to plasmid DNA prior to replication.

State and explain the role of RNAll in plasmid replication. [2]

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[Total: 15]

QUESTION 2

The location of the gene locus responsible for cystic fibrosis was not discovered until 1985. Scientists used restriction fragment length polymorphism to discover the genetic markers associated with the disease. One such marker was a 950bp-long region known as KM19 that had been sequenced prior to 1985.

Samples of DNA were obtained from a family known to have the condition. The KM19 locus was amplified by PCR and mixed with *Pst*I and *Eco*RI in two separate restriction digests. The results of gel electrophoresis of both restriction digests are shown in Fig. 2.1.

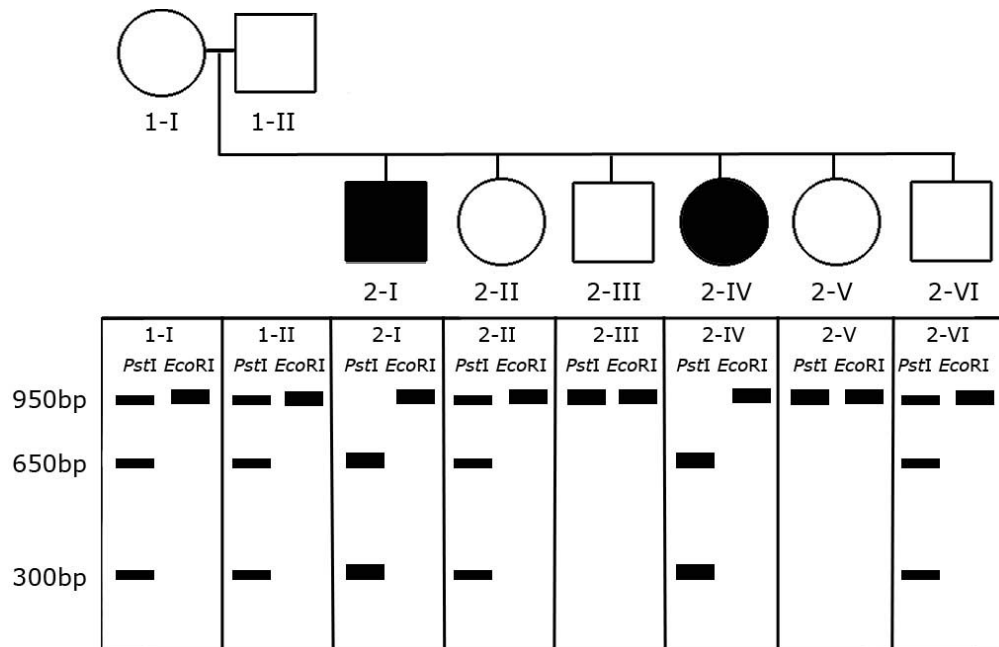


Fig. 2.1

(a) Using the information in Fig. 2.1,

- (i) state and explain which restriction enzyme digest should be used to detect the KM19 genetic marker associated with cystic fibrosis. [3]

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- (ii) draw a restriction map of the KM19 genetic marker that is associated with cystic fibrosis in the space below. [1]

- (b)** Explain why genetic markers like KM19 can be used to detect the presence of disease-causing alleles. [2]

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- (c)** Further research in the 1980s identified sequences of genetic markers located both upstream and downstream of the gene suspected to cause cystic fibrosis.

Using this information, suggest how the gene could be isolated using these two genetic markers. [2]

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Clinical trials for the treatment of inherited conditions via gene therapy have been conducted in the past two decades, but an unsuccessful and lethal trial in 1999 revealed the risks involved in using adenoviruses in gene therapy. Adeno-associated viruses were then used as alternative viral vectors for gene therapy trials as they exhibit little or no disease-causing side effects. More recently, non-viral gene therapies have been explored as safer alternatives to viral vectors.

- (d)** Describe two other advantages of utilising adeno-associated viruses as compared to adenoviruses. [2]

1.

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2.

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- (e)** State two criteria that make cystic fibrosis a condition that can be potentially treated with gene therapy. [2]

1.

2.

Immune responses to non-viral gene therapy have been documented. The main cause of the immune response was found to be the plasmids carrying the normal allele.

- (f) Suggest a reason why plasmids may elicit an immune response during non-viral gene therapy. [1]

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- (g) State three other differences between non-viral gene therapy and viral gene therapy. [3]

1.
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2.
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3.
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[Total: 16]

QUESTION 3

Embryonic stem (ES) and induced pluripotent stem (iPS) cells hold great promise for the treatment of a wide variety of acquired or hereditary diseases. One of the major obstacles to clinical applications is directing them to differentiate into functional cells.

The effect of various growth factors on stem cell differentiation in ectodermic, mesodermic and endodermic embryonic stem cells was carried out and the process was documented in Fig. 3.1. Embryonic stem cells were cultured and embryoid bodies allowed to differentiate into ectodermic, mesodermic and endodermic embryonic stem cells. Growth factors were added to the stem cells and the expression of cell-specific genes was measured after ten days.

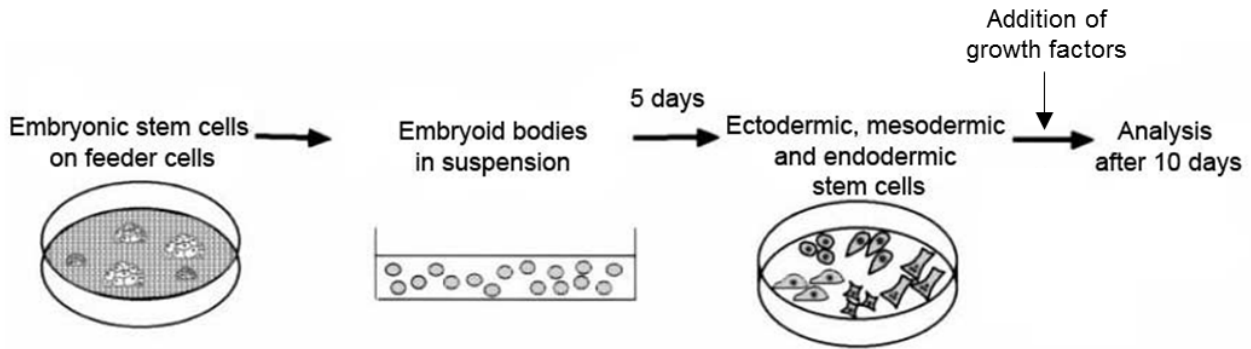


Fig. 3.1

One of the growth factors tested was the epidermal growth factor. 100ng/mL of epidermal growth factor was given to the three different embryonic stem cells. The results are summarised in Table 3.1.

Type of stem cell	Expression of cell-specific genes after ten days		
	Keratin	α -Globin	Insulin
Embryonic Ectoderm	+	—	—
Embryonic Mesoderm	—	+	—
Embryonic Endoderm	—	—	—

+ genes are expressed
— genes are not expressed

Table 3.1

(a) Using the information given and the data from Table 3.1, describe and explain the effect of epidermal growth factor on the expression of keratin in stem cells. [3]

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- (b)** Suggest why the insulin gene was not expressed in the three different kinds of embryonic stem cells even after treatment with a variety of other growth factors. [1]

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Researchers have recently used induced pluripotent stem (iPS) cells generated from human skin cells to create precursor heart cells.

- (c)** State two advantages of using iPS cells instead of embryonic stem cells for research and clinical trials. [2]

1.
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2.
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Cord blood stem cell therapy has been used in clinical trials for the treatment of autoimmune diseases like leukaemia and inherited conditions like Severe Combined Immuno-deficiency (SCID).

The two main modes of treatment are:

- o Allogeneic – The patient receives stem cells from a matching donor, either a sibling or an unrelated donor
- o Autologous – The patient receives their own stem cells

- (d)** Describe the properties of cord blood stem cells. [2]

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- (e)** Suggest a potential limitation of autologous cord blood stem cell therapy. [1]

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[Total: 9]

Question 4 Planning Question

You are required to plan, but not carry out, an investigation into the effect of temperature from 10°C to 70°C on the permeability of potato cell membranes. Potatoes are rich in potassium ions. When small disc-shaped potatoes cut from a core borer (Fig. 4.1) are placed in water, potassium ions are released from the cell into the water.

The concentration of potassium ions in the water can be measured using an electrode (Fig. 4.2) that is selective for potassium ions.



Fig. 4.1



Fig. 4.2

Your plan should:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it.
- Be illustrated by relevant diagram(s) to show, for example, the arrangement of apparatus used.
- Include the layout of result tables and graphs with clear headings and labels.
- Include full details and explanations of the procedure that you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible.

Your planning must be based on the assumption that you have been provided with the following equipment and apparatus which you must use:

- Potatoes
- Potassium ion-selective electrode which measures in mg/L
- A core borer of 10mm in diameter
- Scalpel
- Ruler
- Boiling tubes
- Measuring cylinders
- Stopwatch
- Thermostatically-controlled water bath
- Thermometer
- Isotonic buffer solution
- De-ionized water
- Other common laboratory apparatus

[Total: 12]

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question

QUESTION 5

- (a)** Explain how genetically modified organisms may provide a solution to global food shortage issues. [8]
- (b)** Discuss the social and ethical implications of germ-line gene therapy. [5]
- (c)** Describe the goals of the Human Genome Project. [7]

• End of Paper •